Sidharth V

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.78
Deeksha CFL PU College	Class XII (KSEAB)	2024	98.67%
Deccan International School	Class X (CISCE)	2022	96.30%

SCHOLASTIC ACHIEVEMENTS

Received the **Academic Excellence Award** for scoring the **highest GPA** in my branch

PROJECTS

RISC-V (RV32I) Processor | GitHub

- **Designed and implemented a 32-bit RV32I ISA processor** in Verilog, architecting the complete CPU datapath and control logic, including the ALU, register file, control unit and instruction decoder
- **Upgraded the processor** to a **5-stage pipelined architecture**, implementing **hazard detection**, **data forwarding**, and pipeline control for correct instruction execution
- **Verified architectural correctness** across the RV32I instruction set using Verilog testbenches and RTL simulations

NEGF Resonant Tunneling Diode Simulator | GitHub

- Implemented an RTD simulator with a **self-consistent NEGF-Poisson** solver to model quantum transport
- Constructed **tight-binding Hamiltonian**, **contact self-energies** and **Green's functions** to calculate **electron density** and **transmission/current**
- **Solved nonlinear electrostatics** using Poisson iterations
- Investigated quantum confinement and resonant tunneling behavior

Hybrid Dual-Mode Macropad | GitHub

- **Developed C firmware** using libopencm3 for an STM32F103 MCU, implementing low-level USB HID desktop drivers and nRF24L01 2.4 GHz wireless communication protocols
- **Optimized display refresh rates** for an I2C SSD1306 OLED screen by implementing page-chunking and dirty-bit rendering to minimize bus traffic and latency
- **Developed a custom DFU bootloader** and companion CLI tool to parse JSON configurations, and flash multi-layer keymaps dynamically over USB
- **Collaborated on cross-functional hardware integration**, assisted with 2-layer PCB layout validation, USB-C power routing, and 3D CAD enclosure optimization

SKILLS

Programming languages: C, C++, Python, MATLAB

EDA & PCB: Altium Designer, KiCAD

HDL: Verilog, SystemVerilog, Verilator, Yosys, Icarus Verilog

Simulation: LTspice, NGspice

Tools: Linux, Vim, Git, Make, Docker

Documentation: LaTeX, Typst, Markdown

RELEVANT COURSES

Electronics Courses: Microprocessor and Computer Architecture, Digital Systems, Introduction to HDL, Productivity Tools for IC Design, Analog Electronics, Semiconductor Device Fundamentals, Numerical Methods for Device Modelling, Modelling of Nanoscale MOSFETs, Circuits and Network Analysis, Introduction to CMOS Processing

Programming Courses: Introduction to Programming, Data Structures and Algorithms

Lab: Digital Systems Lab, Semiconductor Devices Lab, Circuits Lab, Analog Lab, Electronic Systems Design Project Lab, PCB Design and Layout

POSITIONS OF RESPONSIBILITY

Teaching Assistant for the **Engineering Electromagnetics** course. My responsibilities covered:

- Creating questions for practice sets and assignments
- Holding doubt clearing sessions for the students

Core member of **Elektronica**, the Electronics and Signals club of IITH.

- Conducted sessions on Computer Architecture and Microprocessors
- Participated in multiple competitions and wrote Medium articles
- Represented the club in the e-Yantra Robotics Competition under the Maze-Solver bot theme

Core member of **Kludge**, the Information Security and networking club of IITH

- Conducted sessions on Linux basics
- Designed a custom P2P file sharing app
- Made challenges for the CTFs held at IITH

EXTRACURRICULAR

Participated in the **e-Yantra robotics competition** 2025-26 hosted by IIT Bombay and funded by the Ministry of Education, Govt. of India. We worked on a Maze solving robot using the DE0-Nano FPGA.